Guidance: Where do I want to go? (Trajectory)

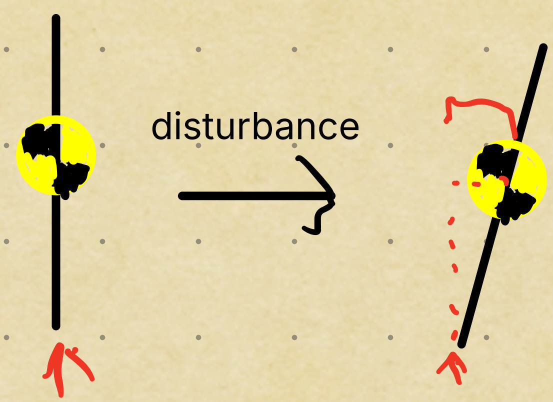
Easy for us: straight up

Navigation: Where am I now? (Sensor Measurements)

Generate *state* of vehicle (position, velocity, etc.)

Control: How do I get there? (Actuators, generate physical command)

Inverted Pendulum: Problem of TVC



Any disturbance leads to a torque that tries to increase itself, making it *unstable.*

Fins make a rocket a regular pendulum, with any disturbance leading to a restoring torque, one that tries to reduce itself.

Intro to PIDs

Proportional: our input is proportional to amount of error (**Understands present)**

Issue: what if we need an output even when error is 0 (e.g. hovering drone)?

This is steady-state error, when system finds equilibrium at != 0 error

Solution: **Integral**: Sums input over time; steady state error will cause integral component to keep on increasing, solving it. (**Remembers past**)

Issue: We may overshoot our target point, causing us to wobble above and below it.

Solution: **Derivative**: Measures rate of error change, slows us down before we overshoot (**Sees the future)**

Each component has a coefficient known as *gain*, we tune the controller by adjusting it.

Sensor fusion:

Using 2 or more data sources to get better picture of reality

4 phases of autonomous system:

-Sense: Collect raw data (noisy electrical signals)

-Perceive: Interpret data (altitude, angular velocity)

Sensor Fusion

-Plan: Find a path (Look at desired trajectory aka guidance)

-Act: Follow path (Calculate error and gimbal nozzle accordingly)

Sensor fusion helps in 4 ways:

-Better reliability (error checking if one instrument fails)

-Noise reduction (averaging outputs of same type reduces random noise by sqrt of number of sensors)

-Different types of sensors also reduces noise, since same type sensors are often noisy in the same way. Done often with a *Kalman Filter*

-We can estimate unmeasured states (e.g. 2 eyes allow depth perception, 1 eye can't).